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(Printed Pages 3)

(20517)

Roll No.

B.Sc. (Micro.)-III Year

3512

B.Sc. (Micro.) Examination, May 2017

Fundamental of Information Technology

(B-309)

Time : Three Hours]

[Maximum Marks : 25 (19)

Note : Attempt any **five** questions. **All** questions carry equal marks.

1. What do you mean by an Operating system? Explain the functions of an Unix operating system. 5
2. What is a topology? Explain Ring topology, Startopology & Bus topology with advantages. 5
3. What is a SQL? Write following Commands:
(a) How to create a new table. 5

P.T.O.

- (b) How to insert the values in a table.
- (c) How to modify the values in a table.
4. Write short notes on any **two** of the following: $2.5 \times 2 = 5$

(a) Functions of an operating system

(b) Anchors & Images

(c) Java Script

5. Differentiate between LAN, MAN & WAN. 5

6. Explain the Object oriented & Relational data base in short. 5

7. What is a Key? Differentiate between Primary key, Secondary key & Foreign key. 5

8. Write the full form of the following: $1 \times 5 = 5$

(a) FTP

(b) SMTP

(c) DNS

(d) HTTP

(e) GUI

9. Write short notes (Any **two**).

(a) Operators

(b) Variables

(c) Components of a DBMS

B. Sc. (Micro.) Examination, May 2018**Fundamental of Information Technology****(B-309)****Time : Three Hours]****[Maximum Marks : 25**

Note : Attempt any *five* questions. All questions carry equal marks.

- ① Write short notes on any two of the following : $2\frac{1}{2} \times 2$
- (a) Relational database model
 - (b) Network database model
 - ~~(c) Hierarchical database model~~
 - ~~(d) Object Oriented database model.~~

- ② What is an operating system ? Explain any two types of operating system in detail.

~~3.~~

What is a key ? Differentiate between primary key and secondary key.

4. Write short notes on any two of the following :

- (a) HTTP
- (b) WWW
- (c) FTP
- (d) Search engine.

5. (a) What is a filter ? Explain when it is used ?

(b) Explain about the pattern matching.

~~6.~~

Differentiate between Ring, Star, Bus and Tree topologies in detail.

~~7.~~

Write down any four SQL commands in detail.

~~8.~~

(a) Explain the functions of an operating system.

(b) Describe the concept of pipes in detail.

Total Questions : 8]

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**B.Sc. (Micro.) IIIrd Year Examination,
May-2019**

**FUNDAMENTAL OF INFORMATION
TECHNOLOGY**

(B-309)

[B.Sc. (Micro.)]

Time : 3 Hrs.]

[M.M. : 25

Note :- Attempt any *five* questions. Each question carries
5 marks.

1. What is an operating system ? Why is it necessary
for a computer system ? List various functions
performed normally by an operating system.

2. Draw logical architecture diagram of a computer system. Explain the role of an operating system in this architecture, name *four* popular operating system.

3. What is network topology ? Describe *three* network topologies in common use. Write their relative advantages and disadvantages.

4. Write short notes on any *two* of the following :

(i) WAN

(ii) LAN

(iii) Ring topology

(iv) BUS topology

5. Write short notes on any *two* of the following :

(i) Hashing

(ii) Arrays

(iii) Operators

(iv) Strings

6. What is SQL ? Explain the following SQL commands :

- (i) Update
- (ii) Modification
- (iii) Deletion
- (iv) Rename

7. Write short notes on any *two* of the following :

- (i) Hierarchical database model
- (ii) Relational database model
- (iii) Network database model
- (iv) Object-oriented database model

8. (i) What is Key ? Explain primary key, secondary and foreign key.

(ii) What is Database ? Explain the advantages and disadvantages of DBMS.

3512 (CV)

B.Sc. (Micro.) IIIrd Year Examination,
June - 2020

FUNDAMENTAL OF INFORMATION
TECHNOLOGY

(B-309)

[B.Sc. (Micro.)]

Time : Two Hours]

[Maximum Marks : 25

Note : Attempt any **four** questions. Each question carries equal marks.

1. What are the different types of operating system? and also Explain the four necessary conditions behind deadlock in operating system.
2. Define network topology. What are the various types of topologies. Explain any two topologies.
3. Differentiate between point to point and multipoint in data communication.

P.T.O.

4. Explain the following terms :
 - (a) Weak entity
 - (b) Strong Entity
 - (c) Primary Key
 - (d) Foreign Key
 - (e) Candidate Key
5. Explain the difference between
 - (a) DELETE and TRUNCATE command
 - (b) SQL and MySQL
6. Define the following terms in PERL
 - (a) Arrays
 - (b) Hashes
 - (c) Variable Context
 - (d) Boolean context
 - (e) Scalar Data types.
7. What are the benefits of PERL Programming in using Web based applications? Explain in detail.
8. What are major features of UNIX operating system that has made it popular? Discuss there salient features of UNIX architecture in brief.

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B.Sc.(Micro.)-III Yr.

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Roll No. 180979131021

3512 (CV-II)

B.Sc. (Micro.) Examination,

June - 2021

Fundamental of Information

Technology

(B-309)

[B.Sc. (Micro.)]

Time : 1½ Hours]

[Maximum Marks : 25

Note : Attempt any **two** questions. Each questions carry equal marks.

1. Why do you think that operating system is essential for computer system? Explain various types of functions performed by an operating system.

P.T.O.

2. Discuss the types of operating systems. How unix or Linux are different from MS-DOS? Also name some unix utilities.
3. Compare and contrast various types of network topologies. Which topology do you think is robust?
4. What do you mean by the terms 'Database' and 'Database Management systems'? What are the roles of various types of keys in a database? Explain with the help of example.
5. Discuss and compare three different types of database models namely Hierarchical, Network and Relational.
6. What do you mean by Structured Query Language (SQL)? Explain various categories of SQL commands. Name few commands also.

7. What do you mean by hypertext? How and where do we use it? Also define scalar data and scalar variable.

8. Short note (Any **two**) :

- (a) Shell programming
- (b) Java script and Perl
- (c) LAN, MAN, WAN
- (d) Routers and bridges

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B.Sc. (Micro.) Examination, June-2022

FUNDAMENTALS OF INFORMATION

TECHNOLOGY

(B-309)

Time : Three Hours] [Maximum Marks : 25

Note : Attempt any **five** questions. Each question carries **5** marks.

1. What are the two primary objectives of having an operating system in a computer system? Explain how an operating system helps in meeting these objectives.
2. Name four popular operating systems. Describe the main features of any two of them.

3. Write short notes on any **two** of the following:
- (a) LAN versus WAN
 - (b) Network Topologies
 - (c) Network connecting devices
 - (d) WAN versus MAN
4. Describe any four network topologies in common use. Write their advantages and disadvantages with examples.
5. What is a database model? Name four commonly used database models. Describe any two.
6. What are four major components of a database management system (DBMS)? Describe roles of each of these components in overall design and usage of a database.
7. What is SQL? How it is useful? Write the "update" command and "Create" Command.

Write short notes on any **two** of the following:

- (a) Strings
- (b) Arrays
- (c) Java script
- (d) Hypertext linking

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B.Sc. (Micro) Examination, May-2024

FUNDAMENTALS OF INFORMATION

TECHNOLOGY

(B-309)

B.Sc. (Micro.)

Time : Three Hours] [Maximum Marks : 25

Note : Attempt any **five** questions. Each question carries **5** marks.

1. What is operating system? Write different services provided by operating system.
2. What are the basic differences between Network operating system and Distributed operating system? Name the different types of operating system.

P.T.O.

3. What are the different types of keys in Database Management system (DBMS).
Why do we require keys in DBMS.
4. What is SQL is database Management system (DBMS). How many SQL statement are used? Enlist some commands of DDL, DML and DCL.
5. Write short notes on any **two** of the following :
- (a) Tree versus BUS Topology.
 - (b) Hub, Switch and Router
 - (c) LAN versus MAN
 - (d) Internet versus Intranet
6. What are the different types of a network? Explain each briefly.
7. Write the syntax of inserting the values in a table and deletion in database.

8. Write short notes on any **two** of the following :

(a) Shell programming

(b) Types of operators used in perl

(c) Pattern Matching

(d) Scalar Variable

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Roll No.

B.Sc. (Micro)-III Year

3512**B.Sc. (Micro.) Examination, April-2025****FUNDAMENTAL OF INFORMATION****TECHNOLOGY****(B-309)****[B.Sc. (Micro.)]*****Time : Three Hours]******[Maximum Marks : 25***

Note : Attempt any **five** questions. Each question carries equal marks.

1. Differentiate among the following types of Operating System by defining their essential properties.
 - (a) Time sharing system
 - (b) Realtime system

(c) Parallel system

(d) Simple batch system

2. Define Operating System. Also explain pipe and filter command in Unix with example.

3. Discuss various types of network topologies in computer network.

Also discuss various advantages and disadvantages of each topology.

4. Write short note on any **two**-

(a) Features of LAN, MAN, WAN

(b) Applications of Computer Networks

(c) Network connecting devices with example

(d) Internet vs Intranet with examples

5. Create a Table named STUDENT with the following structure.

Student-Id	Number
First-Name	Char 20
Last-Name	Char 20
GPA	Float (2,2)
Enrollment-date	Date
Fellowship	Number

- (i) Display the name of the student having the lowest fellowship.
 - (ii) Write an SQL query to fetch the total no. of students.
 - (iii) Write an SQL query to fetch students full names with $GPA \geq 8.5$ and ≤ 9.5
6. Explain DBMS. What is difference between Primary, Secondary and Foreign key.

7. What are the differences between DBMS and RDBMS. Who is the DBA? What are the responsibilities of a DBA?
8. Write short note on any **three** -
- (a) Hashes and Pattern Matching
 - (b) Hypertext linking and forms
 - (c) Operators and Arrays
 - (d) Java Script
 - (e) CGI